

Contracts Module - Microservice Extraction Plan

Public Entry Point

The module exposes a single facade interface: `IContractsService`

- `PrepareContractAsync(PrepareContractRequest, CancellationToken)` - creates a new contract
- `SignContractAsync(Guid contractId, DateTimeOffset signedAt, CancellationToken)` - signs an existing contract

Other modules interact with Contracts only through this interface or by consuming its published events.

API

If extracted as a microservice, the Contracts module would expose:

- `POST /api/contracts` - Prepare a new contract
- `PATCH /api/contracts/{id}` - Sign an existing contract

Data Ownership

- The Contracts microservice would own its own PostgreSQL database (or schema)
- Table: `Contracts` (Id, CustomerId, PreparedAt, Duration, SignedAt, ExpiringAt)
- No other service would have direct access to this data

Module Communication

- Currently: `ContractSignedEvent` is published via an in-memory event bus (MediatR)
- As a microservice: the event would be published to a message broker (e.g., RabbitMQ or Kafka)

- Synchronous calls from other modules to `IContractsService` would be replaced by API calls